STIVEN CATALYST

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# MANAGEMENT REVIEW

Management without theatre.




OPERATIONS

## THE EIGHT WASTES OF LEAN

Ohno's seven wastes, the eighth that came later, five principles, and a walk to find them in your own operation.

**3 h** of value-creating work in the 319 days it took a cola can to reach a buyer.

WOMACK & JONES, 1996

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NO. 05 · OPERATIONS

# In this issue

Most of the time a product or a request spends in a process, nobody is working on it. It waits, it is moved, it is checked again. Learning to see that time is the subject here, along with the names Lean gives it.

Taiichi Ohno, who shaped the Toyota Production System, described seven kinds of waste. Later an eighth was added: the talent of the people that is not used. James Womack and Daniel Jones turned the ideas into five principles in 1996. You do not need a factory for any of it. A warehouse, a delivery route or a hotel reception has the same wastes under other names.

*Stiven Janaqi* EDITOR · STIVEN CATALYST


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## HOW TO READ THIS ISSUE

**Figure**
Every figure has its source and year at the foot of its page.

**Our reading**
Where the editor interprets rather than the research, it says so.

**Practice**
The steps and the card are proposals to try, not research results.

COVER STORY

# 319 days for a cola can

In Lean Thinking, Womack and Jones traced a simple product: a can of cola. It took 319 days, nine facilities, six companies and four countries to reach a buyer.

THE COLA CAN, ONE SQUARE FOR EACH DAY



**319 days**
from the raw material to the buyer

**3 hours**
of value-creating work: one eighth of the last square

*One cola can, traced by Womack and Jones in Lean Thinking (1996): an extreme case, not an average.*

The rest was storage, transport, inspection and rework: steps that create no value for the person who drinks from the can. The can is an extreme case, but the shape is common: a long wait around a short piece of real work.

OUR READING

Before making the work faster, look at the time when no one is working. That is usually where most of the time goes.

Source: James P. Womack & Daniel T. Jones, 1996 (via Lean Enterprise Institute, Seeing the Whole)

THE MODEL

# Seven wastes, and an eighth

The first seven are the ones Ohno described. The eighth was added later; Toyota itself still counts seven.

| | | |
|---|---|---|
| 01 | **Overproduction** | Making more, or earlier, than needed. The worst, because it feeds the others. |
| 02 | **Waiting** | People or machines waiting for material, information or a decision. |
| 03 | **Transport** | Moving material or information more than the work needs. |
| 04 | **Over-processing** | More work than the customer needs or will pay for. |
| 05 | **Inventory** | Material, work in progress or products beyond what is needed now. |
| 06 | **Motion** | Walking, reaching and searching that the task does not need. |
| 07 | **Defects** | Errors that must be corrected, reworked or thrown away. |
| 08 | **Unused talent** | Ideas, skills and knowledge of the people that nobody asks for. |

OUR READING

The eighth is the one a manager controls most directly. The people who do the work usually see the other seven first.

Sources: Taiichi Ohno, 1988; Lean Enterprise Institute, 2014; Lean Enterprise Institute, The Lean Post, 2020; Jeffrey K. Liker, 2004

THE PRINCIPLES

# Five steps of Lean thinking

Womack and Jones summed up Lean as a sequence. Each step starts from the customer, rather than from the machine or the department.

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| **Value** | **Value stream** | **Flow** | **Pull** | **Perfection** |
| What does the customer really pay for? | Every step the product or request goes through. | The steps that add value follow each other without stops. | Make only what the next step or the customer asks for. | Start again: there is always more waste to find. |

HYPOTHETICAL EXAMPLE, A HOTEL ROOM BETWEEN TWO GUESTS

| | |
|---|---|
| **Value** | A clean room, ready when the next guest arrives. |
| **Value stream** | Check-out, cleaning, inspection, room marked ready. Note how long the room waits between steps. |
| **Flow** | Each room is cleaned as soon as the guest leaves, not all of them after noon. |
| **Pull** | The arrivals list sets the order: the rooms needed first are cleaned first. |
| **Perfection** | Each week one cause of delay is removed, such as linen missing from the cart. |

Illustrated by the editors; no real company is described.

**1978**
Ohno's book on the Toyota system appears in Japan.

**1988**
English edition. John Krafcik's article names “lean production”.

**1990**
The Machine That Changed the World: 90 plants in 17 countries.

**1996**
Lean Thinking: the five principles.

Sources: James P. Womack & Daniel T. Jones, 1996 (via Lean Enterprise Institute, Seeing the Whole); Taiichi Ohno, 1988; John F. Krafcik, Sloan Management Review, 1988; James P. Womack, Daniel T. Jones & Daniel Roos, 1990

BEYOND THE FACTORY

# The same wastes, other names

Three settings, a few examples each. They are illustrations, not cases from a particular company: the point is to recognise the pattern in your own work.

### Warehouse

- **01** Picking orders long before the truck is planned
- **06** Walking to a far shelf for the fastest item
- **07** Parcels loaded without the last item

### Last mile

- **02** Drivers waiting at the gate for loading
- **03** A parcel driven twice because the address was unclear
- **05** Returns piling up until the end of the week

### Hotel

- **04** The same guest data typed into two systems
- **02** A guest waiting for a room that is clean but not released
- **08** A fix from the night receptionist that nobody asks about

## Three questions to find it

1. Where does the work wait, and for how long?
2. What is written, counted or checked twice?
3. What does someone search for every day, because it has no fixed place?

OUR READING

A waste that everyone sees every day usually stops looking like waste. It becomes “the way we work”.

The examples are illustrations by the editors; the numbers refer to the wastes on page 4.

More in the essay — Pareto and 5 Why in practice, not in PowerPoint

HOW IT IS MEASURED

# How much of the time creates value?

Divide the time spent on value-creating work by the total time from request to delivery. For the cola can, three hours out of 319 days is about 0.04%.

1. **Follow one item**
   One order, one request, one room, from the moment it is asked for to the moment it is delivered.

2. **Write down every step and its time**
   Including the time it waits between steps.

3. **Mark the value**
   Would the customer pay for this step if they saw it? If not, it is one of the eight.

4. **Start with the biggest wait**
   The step that is easiest to speed up can wait.

HYPOTHETICAL EXAMPLE, ONE ORDER IN A WAREHOUSE

The order arrives on Monday at 9:00 and is delivered on Wednesday at 15:00: 54 hours. The work that creates value, picking, packing and loading, takes 40 minutes: about 1.2%. The biggest wait is the first night in the queue. (The figures are made up.)

OUR READING

Lean calls overproduction the worst waste because it hides the others: when there is always stock, nobody sees the waiting, the defects or the extra moves behind it.

Sources: James P. Womack & Daniel T. Jones, 1996 (via Lean Enterprise Institute, Seeing the Whole); Lean Enterprise Institute, 2014

The 0.04% is the editors' calculation from the figures of Womack and Jones. The four steps are the editors' suggestion.

TOOL OF THE ISSUE

# The waste walk card

Thirty minutes on the floor, with the people who do the work. One line for each waste: what you saw, where, and how often.

**01 Overproduction** made before anyone asked?

**02 Waiting** who waits for what?

**03 Transport** what is moved more than once?

**04 Over-processing** which step would the customer not pay for?

**05 Inventory** what piles up, and where?

**06 Motion** who walks or searches without need?

**07 Defects** what is done twice?

**08 Unused talent** which idea from the team is waiting for an answer?

The editors propose this practice. Note what you see, not who did it.

Open the tool: Pareto 80/20 stivencatalyst.com/tools/pareto/

SOURCES AND METHOD

# Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.


**Lean Thinking**
James P. Womack & Daniel T. Jones, 1996 · via Lean Enterprise Institute, Seeing the Whole
https://www.lean.org/wp-content/uploads/2021/01/seeing-the-whole-introduction.pdf

**Toyota Production System: Beyond Large-Scale Production**
Taiichi Ohno, 1988

**Lean Lexicon: seven wastes**
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/seven-wastes/

**The Eight Wastes of Lean**
Lean Enterprise Institute, The Lean Post, 2020
https://www.lean.org/the-lean-post/articles/the-eight-wastes-of-lean/

**The Toyota Way**
Jeffrey K. Liker, 2004

**Triumph of the Lean Production System**
John F. Krafcik, Sloan Management Review, 1988
https://www.lean.org/downloads/MITSloan.pdf

**The Machine That Changed the World**
James P. Womack, Daniel T. Jones & Daniel Roos, 1990


**EDITORIAL METHOD**

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



# MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

## ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html